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Pertemuan 2 MA4171
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Fungsi Transfer dan Diagram Blok

MA4171 Teori Kontrol Linear

Soal Latihan

- Tentukan solusi dari $y'' - y' - 2y = 0$ dengan $y(0) = 1, y'(0) = 0$
- Tentukan solusi dari $y'' + y = \sin 2t$ dengan $y(0) = 2, y'(0) = 1$
- Tentukan solusi dari $y^{(4)} - y = 0$ dengan $y(0) = 0, y'(0) = 1, y''(0) = 0, y'''(0) = 0$

(2) Diberikan, $y'' + y = \sin(2t)$, $y(0) = 2$, $y'(0) = 1$

Transformasi Laplace

$$\mathcal{L}\{y'' + y\} = \mathcal{L}\{\sin(2t)\}$$

$$[s^2 Y(s) - sy(0) - y'(0)] + Y(s) = \frac{2}{s^2 + 4}$$

$$s^2 Y(s) - 2s - 1 + Y(s) = \frac{2}{s^2 + 4}$$

$$Y(s)[s^2 + 1] = 2s + 1 + \frac{2}{s^2 + 4}$$

$$Y(s) = \frac{2s+1}{s^2+1} + \frac{2}{(s^2+1)(s^2+4)} = \frac{(2s+1)(s^2+4)+2}{(s^2+1)(s^2+4)}$$

Pandang, $Y(s) = \frac{(2s+1)(s^2+4)+2}{(s^2+1)(s^2+4)} = \frac{2s+1}{s^2+1} + \frac{2/3}{s^2+1} + \frac{-2/3}{s^2+4}$

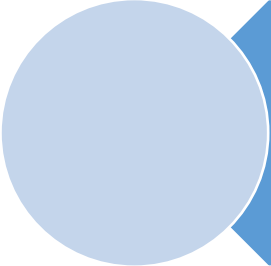
$$= \frac{2s}{s^2+1} + \frac{5/3}{s^2+1} - \frac{2/3}{s^2+4} = 2 \frac{s}{s^2+1} + \frac{5}{3} \cdot \frac{1}{s^2+1} - \frac{1}{3} \cdot \frac{2}{s^2+4}$$

Invers Transf. Laplace

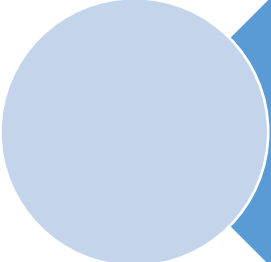
$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left[2 \frac{s}{s^2+1} + \frac{5}{3} \cdot \frac{1}{s^2+1} - \frac{1}{3} \cdot \frac{2}{s^2+4}\right]$$

$$y(t) = 2 \cos(t) + \frac{5}{3} \sin(t) - \frac{1}{3} \sin(2t)$$

Sistem dapat di representasikan dalam bentuk



Persamaan diferensial (kontinu) atau persamaan beda (diskrit)



Fungsi transfer

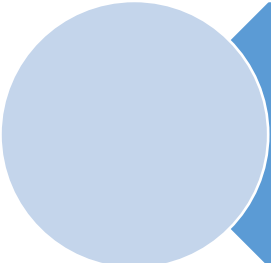
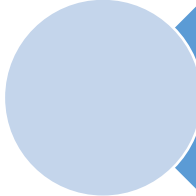
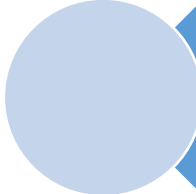


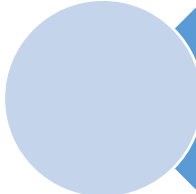
Diagram blok



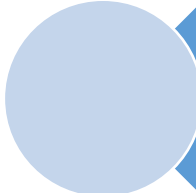
Satu masukan dan satu keluaran (SISO)



Satu masukan dan banyak luaran (SIMO)



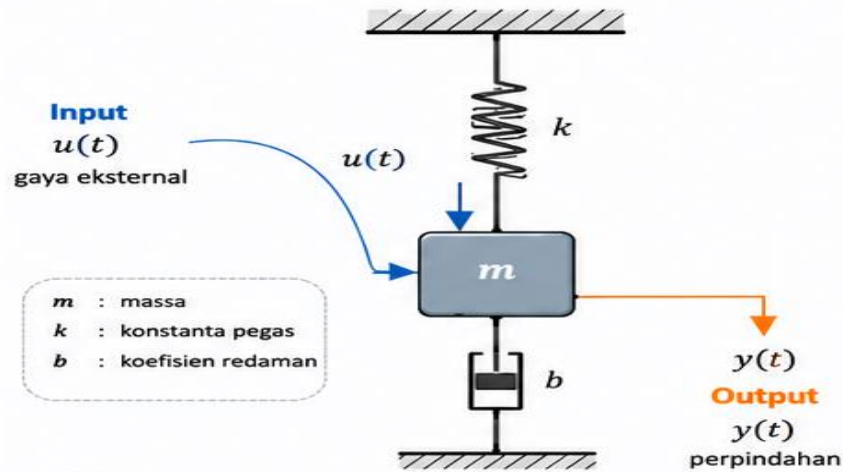
Banyak masukan dan satu luaran (MISO)



Banyak masukan dan banyak luaran (MIMO)

Single Input Single Output (SISO)

Satu masukan menghasilkan satu keluaran.



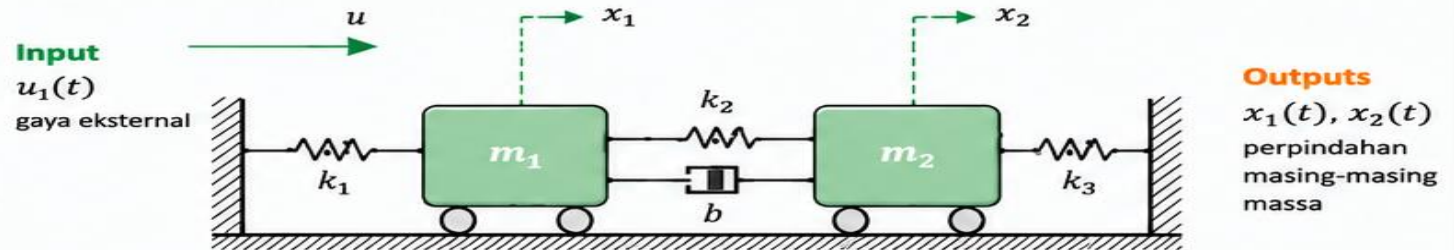
Contoh: Sistem suspensi kendaraan

Input: Gaya dari jalan

Output: Perpindahan bodi mobil

Single Input Multi Output (SIMO)

Satu masukan menghasilkan beberapa keluaran.



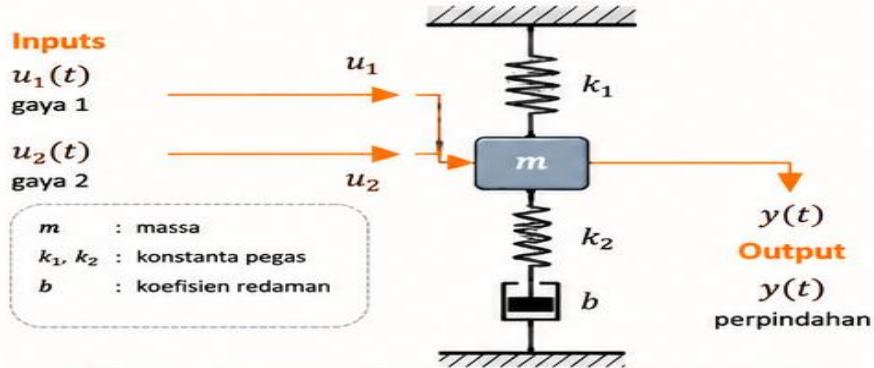
Contoh: Posisi dua massa pada sistem peredam

Input: Gaya eksternal

Outputs: Perpindahan massa m_1 dan m_2

Multi Input Single Output (MISO)

Beberapa masukan menghasilkan satu keluaran.



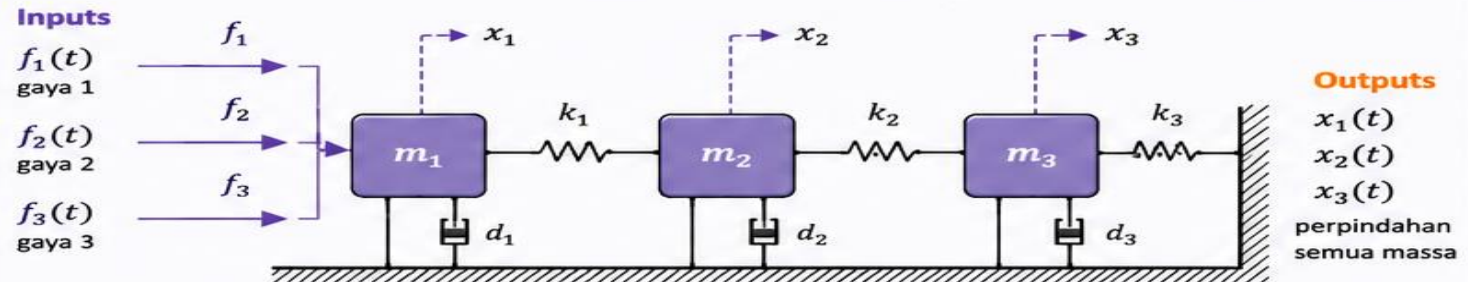
Contoh: Sistem bangunan dengan dua beban

Inputs: Beban angin (u_1) dan beban gempa (u_2)

Output: Perpindahan lantai bangunan

Multi Input Multi Output (MIMO)

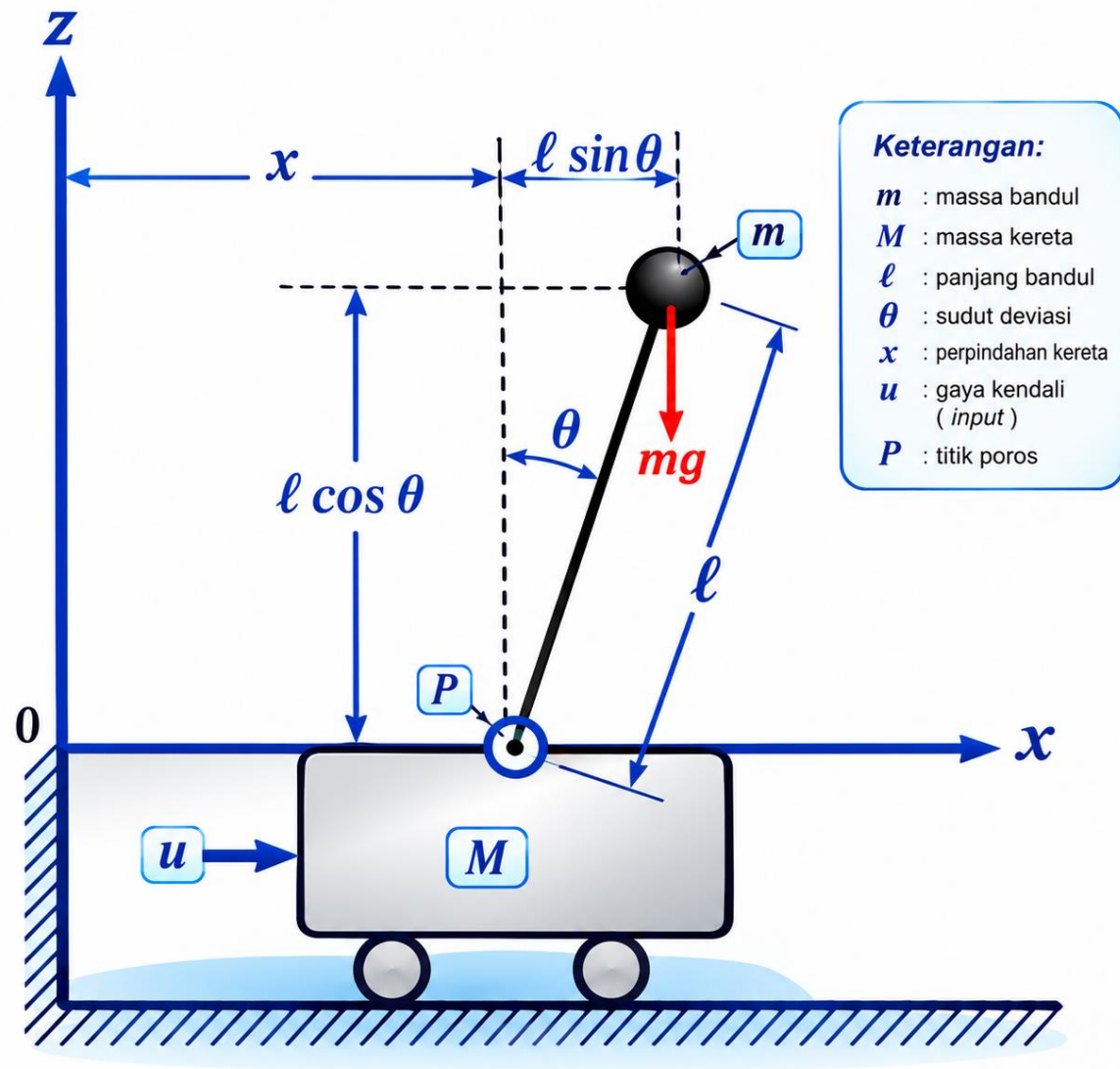
Beberapa masukan menghasilkan beberapa keluaran.



Contoh: Sistem struktur bangunan bertingkat

Inputs: Beban angin, beban gempa, dan beban manusia

Outputs: Perpindahan setiap lantai



$$x_1 = \theta$$

$$x_2 = \dot{\theta}$$

$$x_3 = x$$

$$x_4 = \dot{x}$$

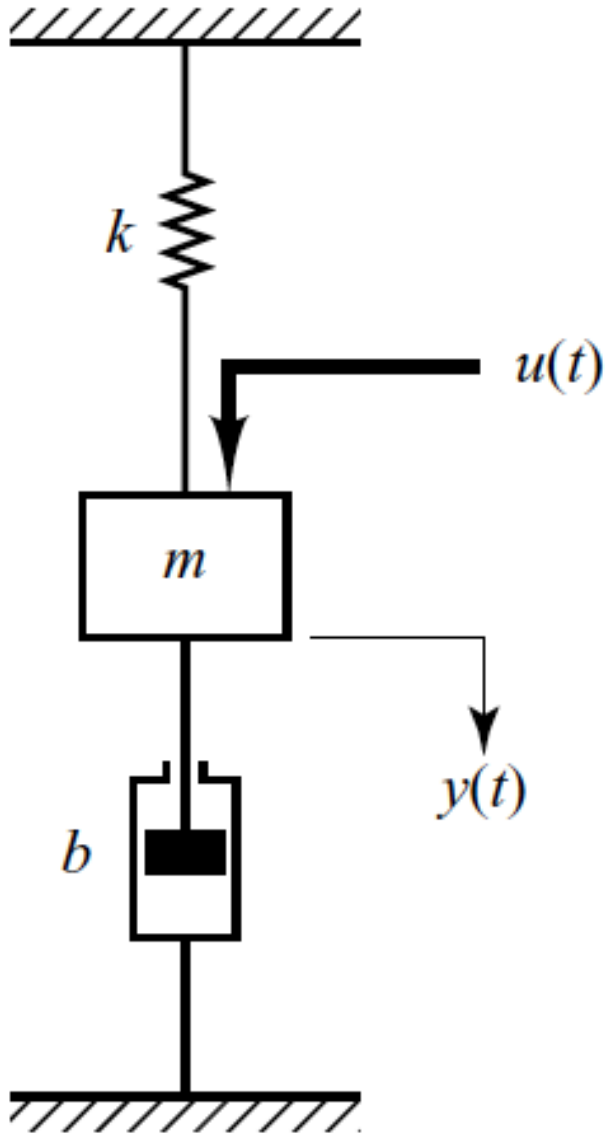
$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{M+m}{Ml}g & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -\frac{m}{M}g & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{1}{Ml} \\ 0 \\ \frac{1}{M} \end{bmatrix} u$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

Fungsi transfer

- Fungsi transfer dari sistem linear dalam persamaan diferensial biasa, time-invariant didefinisikan sebagai ratio transformasi Laplace dari keluaran (output) dan transformasi Laplace dari masukan (input) dengan asumsi semua kondisi awal adalah nol.
- Fungsi transfer = $G(s) = \frac{\mathcal{L}[output]}{\mathcal{L}[input]}$ dengan kondisi awal sama dengan nol.
- Untuk kasus sistem diskrit (persamaan beda, linear, time-invariant), gunakan transformasi Z.

Contoh



Model: $my''(t) + by'(t) + ky(t) = u(t)$

Fungsi transfer:

$$ms^2Y(s) + bsY(s) + kY(s) = U(s)$$

$$(ms^2 + bs + k)Y(s) = U(s)$$

$$Y(s) = \frac{1}{ms^2 + bs + k} U(s)$$

Jadi fungsi transfer: $G(s) = \frac{1}{ms^2 + bs + k}$

Untuk sistem linear orde- n :

$$a_0 y^{(n)}(t) + a_1 y^{(n-1)}(t) + \cdots + a_{n-1} y'(t) + a_n y(t) = \\ b_0 x^{(m)}(t) + b_1 x^{(m-1)}(t) + \cdots + b_{m-1} x'(t) + b_m x(t), \quad n \geq m$$

Fungsi transfernya dapat ditulis

$$G(s) = \frac{b_0 s^m + b_1 s^{m-1} + \cdots + b_m}{a_0 s^n + a_1 s^{n-1} + \cdots + a_n}$$

Latihan:

Tentukanlah fungsi transfer dari sistem berikut.

$$1. y'' + 3y' + 7y = 6u$$

$$2. y''' + 6y'' + 2y' + 4y = -5u'' + 8u'$$

$$3. \begin{cases} x_1' = -3x_1 + x_2 - 4u \\ x_2' = -2x_1 - u \\ y = x_1 - 2x_2 \end{cases}$$

$$4. y'' + 3y' + 7y = 6u_1 + 5u_2 - 4u_2'$$

$$5. \begin{cases} y_1'' + 6y_1' + 2y_1 = u' - 3u \\ y_2' + 6y_2 = 4u' \end{cases}$$

$$6. \begin{cases} y_1''' + 7y_1'' + 6y_1' + y_1 = u_1'' + 3u_1 + u_2 \\ y_2''' + 7y_2'' + 6y_2' + y_2 = 4u_2' \end{cases}$$

$$7. \begin{cases} y_1' + 2y_1 = u_1 + 5u_2 \\ y_1 + y_2' + 3y_2 = 4u_2 + u_2' \end{cases}$$

$$(3) \begin{cases} \dot{x}_1 = -3x_1 + x_2 - 4u \\ \dot{x}_2 = -2x_1 - u \\ y = x_1 - 2x_2 \end{cases}$$

Transformasi Laplace :

$$(i) sX_1(s) = -3X_1(s) + X_2(s) - 4U(s)$$

$$(ii) sX_2(s) = -2X_1(s) - U(s)$$

$$(iii) Y(s) = X_1(s) - 2X_2(s)$$

Menentukan $X_1(s)$ dan $X_2(s)$

$$(a) (s+3)X_1(s) - X_2(s) = -4U(s)$$

$$(b) 2X_1(s) + sX_2(s) = -U(s)$$

Maka,

$$Y(s) = \frac{-4s-1}{s^2+3s+2} U(s) - 2 \cdot \frac{s-5}{-(s^2+3s+2)} U(s)$$

$$= \frac{-2s-11}{s^2+3s+2} U(s)$$

$$\text{Fungsi Transfer : } \frac{Y(s)}{U(s)} = \frac{-2s-11}{s^2+3s+2}$$

• Kalikan (a) dgn s :

$$s(s+3)X_1(s) - sX_2(s) = -4sU(s)$$

$$2X_1(s) + sX_2(s) = -U(s) \quad +$$

$$(s(s+3)+2)X_1(s) = (-4s-1)U(s)$$

$$X_1(s) = \frac{-4s-1}{s^2+3s+2} U(s)$$

• Kalikan (a) dgn 2 dan kalikan (b) dgn $(s+3)$

$$2(s+3)X_1(s) - 2X_2(s) = -8U(s)$$

$$2(s+3)X_1(s) + s(s+3)X_2(s) = -(s+3)U(s) \quad -$$

$$-2X_2(s) - s(s+3)X_2(s) = (s-5)U(s)$$

$$-(s^2+3s+2)X_2(s) = (s-5)U(s)$$

$$X_2(s) = \frac{s-5}{-(s^2+3s+2)} U(s)$$

Representasi Sistem Dinamik

Menuliskan representasi system dinamik melalui fungsi transfer dan diagram blok

Model dalam bentuk persamaan ruang keadaan

- Model dalam bentuk time-variant:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t)$$

$$y(t) = C(t)x(t) + D(t)u(t)$$

- Model dalam bentuk time-invariant:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

- x : variabel keadaan (state variable), A : matriks keadaan (state matrix),
 u : kontrol masukan (input control), B : matriks masukan (input matrix),
 y : luaran (output), C : matriks luaran (output matrix).

- Model dalam bentuk time-invariant:

$$\dot{x}(t) = Ax(t) + Bu(t)$$

$$y(t) = Cx(t) + Du(t)$$

- $Y(s) = G(s)U(s)$
- $G(s) = C(sI - A)^{-1}B + D$

Contoh:

- Sistem massa pegas dengan model $my''(t) + by'(t) + ky(t) = u(t)$.
Dengan mendefinisikan $y = x_1$, dan $y' = x_2$, persamaan ruang keadaannya dapat ditulis dalam bentuk:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & -\frac{b}{m} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} u, \quad y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Contoh

(1) Tentukan model linear dari persamaan $\ddot{y} + 3\ddot{y} + 2\dot{y} = u$

(2) Tentukan fungsi transfer dari system berikut:

$$\dot{x}_1 = -4x_1 - x_2 + u$$

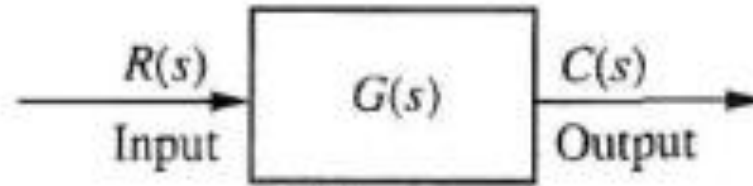
$$\dot{x}_2 = 3x_1 - x_2 + u$$

$$y = x_1$$

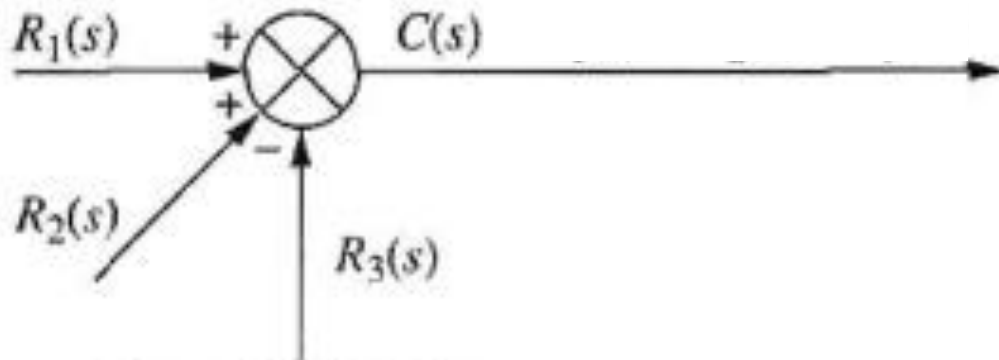
Diagram Blok: menggambarkan aliran sinyal



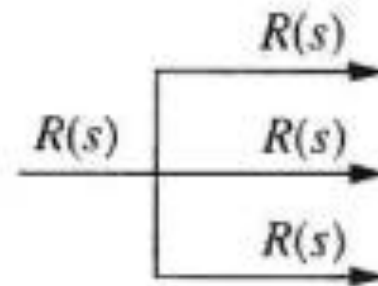
Signals
(a)



System
(b)

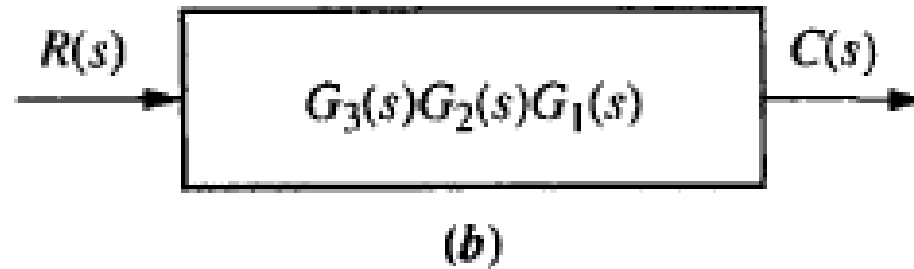
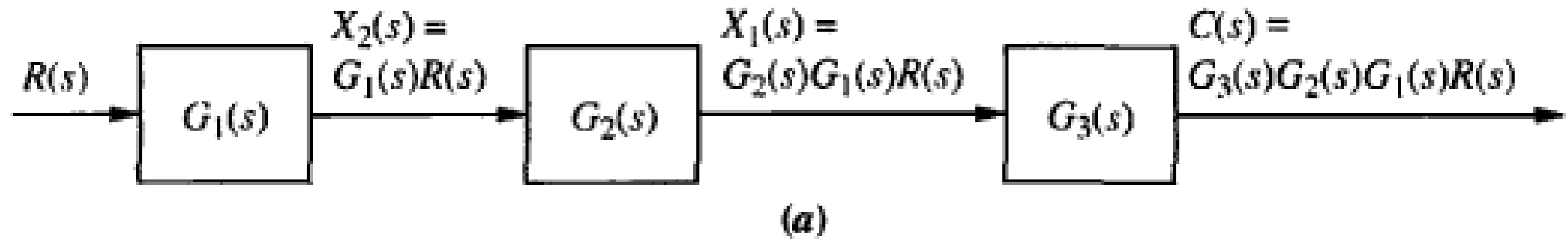


Summing junction
(c)



Pickoff point
(d)

Cascade Form



Parallel Form

